

§12.4 The Cross Product

Homework : 5,9,15,23,27,31,34

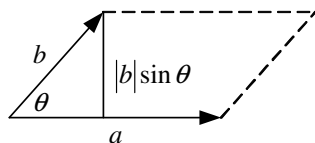
Cross Product (a vector) $a \times b$, $a = \langle a_1, a_2, a_3 \rangle$, $b = \langle b_1, b_2, b_3 \rangle$.

(1)

$$(i) \text{ 代數意義 : } a \times b = \begin{pmatrix} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}, \begin{vmatrix} a_3 & a_1 \\ b_3 & b_1 \end{vmatrix}, \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \end{pmatrix} = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

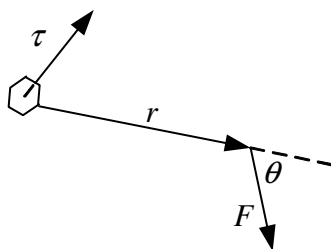
$$i = \langle 1, 0, 0 \rangle , j = \langle 0, 1, 0 \rangle , k = \langle 0, 0, 1 \rangle$$

$$(ii) \text{ 幾何意義 : } a \times b = \begin{cases} \text{大小 : } |a||b|\sin\theta . \\ \text{方向 : right hand rule .} \end{cases}$$



$$\Rightarrow |a \times b| = |a||b|\sin\theta = \text{the parallelogram determined by } a \text{ and } b .$$

(iii) 物理意義 : a force F acting on a rigid body at a point given by a position vector r . Then the torque τ is defined to be $\tau = r \times F$.

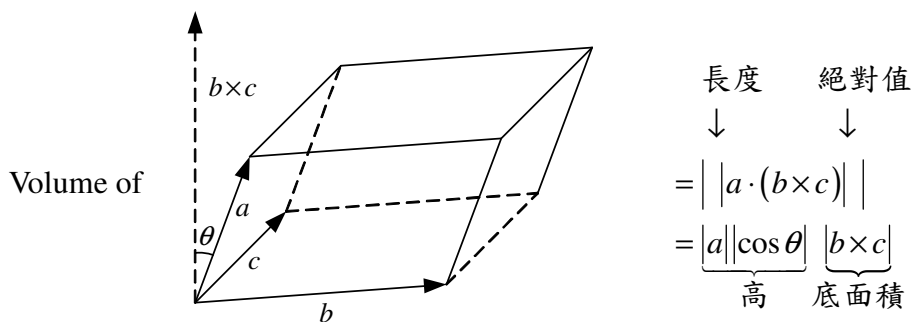


$$(2) a \times b = 0 \Leftrightarrow a \parallel b .$$

$$(3) a \times b \perp a , a \times b \perp b .$$

$$(4) \text{ Area of } \begin{array}{c} \text{parallelogram} \\ \text{determined by } a \text{ and } b \end{array} = |a||b|\sin\theta = |a \times b| .$$

(5)



(6)

(i) $a \times b = -b \times a$.

(ii) $a \times (b + c) = a \times b + a \times c$.

(iii) $a \cdot (b \times c) = (a \times b) \cdot c$

(iv) $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c \neq (a \times b) \times c$.

$$i \times (i \times j) = i \times k = -j$$

$$(i \times i) \times j = 0 \times j = 0$$

Example 1 : Find the area of ΔPQR ,

where $P(1, 4, 6)$, $Q(-2, 5, -1)$ and $R(1, -1, 1)$.

Solution :

$$\overrightarrow{PQ} = \langle -3, 1, -7 \rangle, \overrightarrow{PR} = \langle 0, -5, -5 \rangle$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \langle -40, -15, 15 \rangle$$

$$= \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}| = \frac{5}{2} \sqrt{81}$$

Example 2 : Are $a = \langle 1, 4, -7 \rangle$, $b = \langle 2, -1, 4 \rangle$ and $c = \langle 0, -9, 18 \rangle$ coplanar?

Solution :

$$\begin{aligned} & \langle 1, 4, -7 \rangle \cdot (\langle 2, -1, 4 \rangle \times \langle 0, -9, 18 \rangle) \\ &= \langle 1, 4, -7 \rangle \cdot \langle 18, -36, -18 \rangle = 0 \\ &\Rightarrow \text{Coplanar (共面)}. \end{aligned}$$